

PROGRAMMING METHODS - LABORATORY 6

Program 01

Given a non-empty two-dimensional array of integers with dimensions $n \times m$, where n and m are natural numbers greater than 0. Every row and every column of this array is sorted in non-decreasing order. Write a program that searches for a given element in the array using four different functions:

- a recursive function determining the first occurrence of the element;
- a recursive function determining the last occurrence of the element;
- an iterative function determining the first occurrence of the element;
- an iterative function determining the last occurrence of the element.

All functions should have average time complexity significantly lower than quadratic, i.e. less than $O(\max(n,m)^2)$.

The first occurrence means the earliest lexicographic occurrence: first the earliest row, and then the earliest column. The last occurrence means the latest lexicographic occurrence. The recursive functions must not contain any loops. The iterative functions must not call another function created by the programmer.

Input

Data are read from standard input. The first value is a positive integer specifying the number of data sets. The data sets follow.

Each data set contains two positive integers from 1 to 100 specifying the number of rows and columns, followed by the array rows according to those dimensions. The data in each set are integers from -2^{15} to $+2^{15}$. At the very end of each data set, a value is provided that must be searched in the array by all four functions.

Output

For each data set, the program prints four lines - the results of the four functions - and the fifth line "---".

```
RecFirst: k at (i1,j1)
RecLast:  k at (i2,j2)
IterFirst: k at (i3,j3)
IterLast:  k at (i4,j4)
---
```

The identifier before the colon indicates the function called. k is the searched value, and i and j are the row and column indices of the found occurrence. If the element is missing, after the colon print: "there is no k ", where k is the searched element.

Example data

Input

```
3
3 4
10 10 10 10
10 20 20 30
20 20 20 40
20
3 4
10 10 10 10
10 20 20 30
20 20 20 40
50
2 3
10 10 10
10 20 20
10
```

Output

```
RecFirst: 20 at (1,1)
RecLast:  20 at (2,2)
IterFirst: 20 at (1,1)
IterLast:  20 at (2,2)
```

```
---
RecFirst: there is no 50
RecLast: there is no 50
IterFirst: there is no 50
IterLast: there is no 50
---
RecFirst: 10 at (0,0)
RecLast: 10 at (1,0)
IterFirst: 10 at (0,0)
IterLast: 10 at (1,0)
---
```

Notes

- The first line of the program should contain the comment: // Surname and first name
- Attach a text file with your own test data examples that check all cases appearing in the program and on which the program was run before submission.
- Remember to divide the program into appropriate files and to add comments to the code.
- Protect the program so that it handles exceptions.

Submission rules

The .cpp file(s) should be named:

Surname_Name_Program_01.cpp

The source files, all other created files, and test text files should be placed in the folder Surname_Name_Laboratory_6.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 6 on delta.pk.edu.pl.